

Assignment No. 7

Group 15

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Problem Statement

Prepare a detailed software project plan for the development of a Secure P2P Wallet System. The project plan should include scheduling techniques such as Gantt charts and PERT charts to visualize the development timeline and task dependencies.

Abstract

Software project planning is a critical phase in the software development lifecycle that ensures efficient utilization of resources, proper scheduling of development activities, and successful completion of the project within the defined constraints. Project planning involves identifying tasks, estimating time, defining dependencies, and monitoring project progress.

In this assignment, a project plan for the Secure P2P Wallet System is developed using modern project management techniques. The plan includes task breakdown, scheduling using Gantt charts, and dependency analysis using PERT charts. These techniques help visualize the project workflow and ensure systematic execution of the development process.

Introduction

In modern software engineering, project planning plays a vital role in ensuring the successful development and deployment of complex software systems. Without proper planning, projects may suffer from delays, poor resource management, and cost overruns.

The Secure P2P Wallet System is designed to enable users to perform secure digital financial transactions such as sending money, receiving funds, viewing transaction history, and managing wallet balances. Developing such a system requires careful planning and organization of tasks across multiple development phases.

Project management tools such as Gantt charts and PERT charts provide effective methods for visualizing project schedules and understanding task relationships. These tools allow project managers to track progress, identify potential bottlenecks, and ensure that all project milestones are achieved on time.

Objectives of Project Planning

The primary objectives of software project planning include:

- Identifying all major tasks involved in the project development process.
- Estimating the duration required to complete each task.
- Determining dependencies between tasks.
- Creating a clear project schedule.
- Monitoring project progress and identifying potential risks.
- Ensuring efficient utilization of development resources.

These objectives help maintain a structured development workflow and improve the overall efficiency of the software development lifecycle.

Software Project Planning

The Secure P2P Wallet System development process consists of several stages that collectively contribute to building a reliable and secure financial application. The project planning process begins with identifying the system requirements and continues through design, implementation, testing, and deployment.

The major phases involved in the development of the system include:

- Requirement Analysis
- System Design
- Database Design
- Backend Development
- Frontend Development

- API Integration
- Testing
- Deployment

Each phase represents a critical milestone in the project lifecycle. Proper scheduling ensures that these phases are executed sequentially and efficiently.

Work Breakdown Structure (WBS)

The Work Breakdown Structure (WBS) divides the entire project into smaller manageable components. This hierarchical decomposition helps in organizing project tasks and assigning responsibilities effectively.

Task ID	Task Name	Estimated Duration
T1	Requirement Analysis	1 Week
T2	System Design	1 Week
T3	Database Design	1 Week
T4	Backend Development	2 Weeks
T5	Frontend Development	2 Weeks
T6	API Integration	1 Week
T7	Testing and Debugging	1 Week
T8	Deployment and Documentation	1 Week

Explanation of Tasks

Requirement Analysis: This phase involves gathering functional and non-functional requirements of the system. The development team identifies user needs and defines system specifications.

System Design: During this phase, the architecture of the application is designed. This includes defining system components, database architecture, and interaction between modules.

Database Design: The database schema is created to store user accounts, wallet balances, and transaction history. Proper database design ensures data consistency and integrity.

Backend Development: Backend development involves implementing APIs, business logic, authentication mechanisms, and transaction processing functionality.

Frontend Development: The user interface of the wallet system is developed during this phase. It includes building dashboards, transaction pages, and authentication interfaces.

API Integration: Frontend components are integrated with backend APIs to enable real-time communication between the user interface and server.

Testing and Debugging: Various testing methods such as unit testing, integration testing, and system testing are performed to identify and fix bugs.

Deployment: The final phase involves deploying the application to production servers using containerization technologies such as Docker and Kubernetes.

Project Scheduling Strategy

Project scheduling is the process of defining the timeline for each development task. Scheduling helps ensure that tasks are completed within the allocated timeframe and that project milestones are achieved.

Two important tools used for project scheduling are:

- Gantt Chart
- PERT Chart

These tools help project managers visualize timelines and understand task dependencies.

Gantt Chart

A Gantt chart is a graphical representation of project activities plotted against time. It helps visualize the start and end dates of each task and provides a clear view of the project schedule.

Analysis of the Gantt Chart

The Gantt chart illustrates the timeline of the project from the requirement analysis phase to the final deployment phase. Each task is allocated a specific duration, ensuring that development progresses in a structured manner.

The chart helps project managers track task completion and detect delays early. It also enables efficient coordination among development team members.

Secure P2P Wallet System Development Schedule

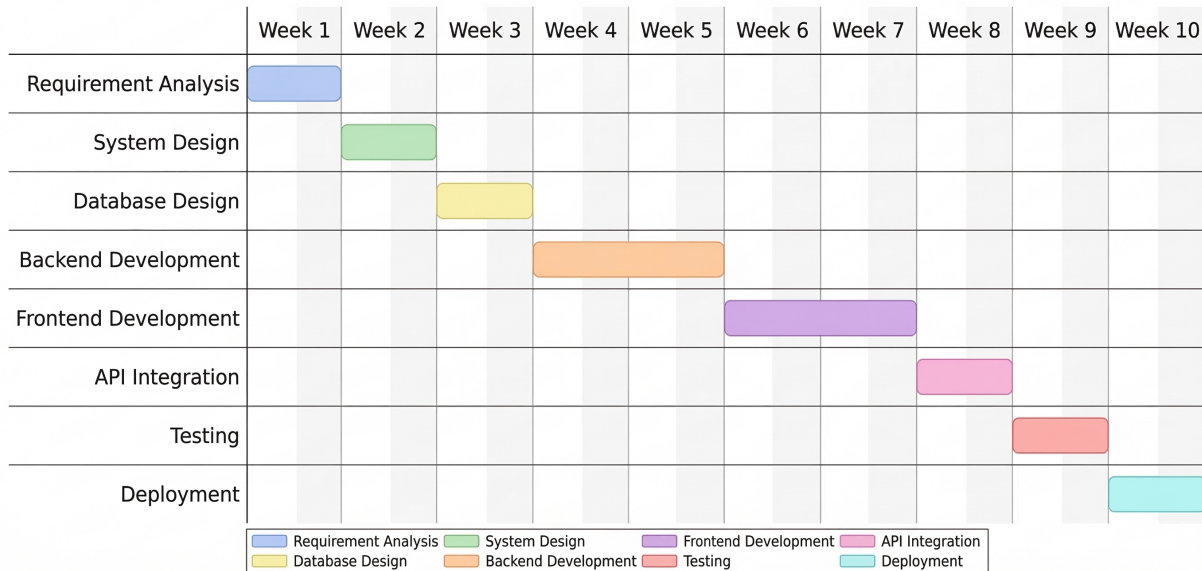


Figure 1: Gantt Chart for Secure P2P Wallet Project Scheduling

PERT Chart

Program Evaluation and Review Technique (PERT) is a project management tool used to analyze task dependencies and determine the sequence of project activities.

Analysis of the PERT Chart

The PERT chart represents project activities as nodes connected by arrows that indicate dependencies between tasks. It helps determine the order in which tasks must be performed and highlights the critical path.

The critical path represents the longest sequence of dependent tasks that determines the minimum time required to complete the project.

Critical Path Analysis

The critical path for the Secure P2P Wallet System development includes the following tasks:

- Requirement Analysis
- System Design

Secure P2P Wallet System Development Workflow

Secure P2P Wallet System Development Workflow

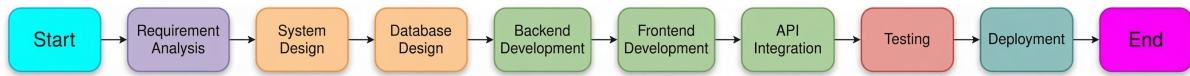


Figure 2: PERT Chart Showing Task Dependencies

- Database Design
- Backend Development
- Frontend Development
- API Integration
- Testing
- Deployment

Any delay in these tasks may affect the overall project completion time. Therefore, careful monitoring of these tasks is essential.

Project Risk Considerations

Several risks may affect the development process of the wallet system:

- Delays in requirement gathering
- Integration challenges between frontend and backend

- Security vulnerabilities in financial transactions
- Performance issues under heavy load
- Deployment and infrastructure challenges

Proper project planning and continuous monitoring help mitigate these risks.

Conclusion

Software project planning is a crucial step in ensuring the successful development of complex software systems. The use of scheduling techniques such as Gantt charts and PERT charts provides valuable insights into project timelines and task dependencies.

In this assignment, a detailed project plan for the Secure P2P Wallet System was prepared. The Work Breakdown Structure defined the major development tasks, the Gantt chart illustrated the project timeline, and the PERT chart demonstrated task dependencies and workflow.

These project management tools help development teams organize their work effectively, monitor project progress, and ensure timely completion of the software system.